

SciHarnessBench: A Self-Generating Benchmark for Fake Science in AI Scientific Agents

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Abstract

Autonomous AI agents increasingly carry out scientific analyses end to end, yet existing benchmarks largely measure whether a model *knows* science or whether it can *complete* a workflow—not whether it catches the data flaws that turn a plausible workflow into a wrong conclusion. We call this failure mode *fake science*: a confident, well-formatted, and incorrect result produced because the agent trusted flawed data. We introduce SciHarnessBench, a benchmark that measures fake science directly. Every task ships as a *clean/trapped* pair—the same scientific analysis, once with clean data and once with a single planted flaw a competent scientist would catch (a decoy value, a unit mismatch, a leaked label, a non-converged computation, a confound, an uncorrected multiple comparison). The headline metric is the Fake-Science Gap, the drop in accuracy from the clean to the trapped twin. SciHarnessBench is *procedurally instantiated* from human-authored templates and *deterministically self-graded*: ground truth is computed by real scientific libraries and detection is verified against it, so there is no human annotation at evaluation time and no LLM judge. The design is contamination-resistant (instances regenerate under a private salt), opaque (the agent never sees the variant), and gaming-resistant (flagging or abstaining on a clean task is a false alarm that scores zero; detection requires structured, evidence-bearing flaw reports). v1 spans 8 domains, 26 task families, and all 12 trap types. Two reference agents validate the design: a careful agent that validates inputs passes clean and trapped tasks (gap 0.0 pts), while a naive agent that trusts inputs solves clean tasks but fails essentially every trap (gap 100.0 pts; 92.4 % of failures are confident-wrong), confirming that SciHarnessBench isolates fake science with a wide, well-separated dynamic range. We release the benchmark, harness, tests, and evaluation protocol.

1 Introduction

AI systems are moving from answering scientific questions to *conducting* science: writing and running analysis code, calling domain tools, and reporting conclusions with little human oversight. The central question for trusting such an agent is not whether it can produce an analysis, but whether the analysis is *right*—and, crucially, whether the agent notices when the data make the obvious analysis wrong. A scientist’s value lies as much in skepticism as in computation:

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checking units, validating inputs, confirming a solver converged, guarding against confounds and leakage. An agent that skips these steps can produce output that looks impeccable and is completely wrong. We call this *fake science*.

Most AI-for-science benchmarks do not measure it. Knowledge benchmarks such as GPQA (Rein et al. 2024) and ChemBench (Mirza et al. 2025) test what a model knows. Predictive benchmarks such as MoleculeNet (Wu et al. 2018), PDEBench (Takamoto et al. 2022), WeatherBench 2 (Rasp et al. 2024), ProteinGym (Notin et al. 2023), and Matbench Discovery (Riebesell et al. 2025) test predictive accuracy on fixed tasks. Agentic benchmarks such as ScienceAgentBench (Chen et al. 2025), MLE-bench (Chan et al. 2025), BixBench (Mitchener et al. 2025), DiscoveryWorld (Jansen et al. 2024), LAB-Bench (Laurent et al. 2024), and AstaBench (Bragg et al. 2026) test whether an agent can *complete* a data-driven workflow. All are valuable, and none directly asks the question that predicts trust: *when the data hide a flaw, does the agent catch it or confidently report a wrong result?*

We introduce SciHarnessBench to measure exactly this. Its core construct is the *clean/trapped* pair: a scientific task instantiated twice from the same template and seed, identical in every respect except that the trapped twin contains one planted flaw drawn from a taxonomy of 12 documented, machine-detectable failure modes (Section 5). An agent’s *competence* is its accuracy on clean tasks; its *robustness* is its accuracy on trapped tasks; the Fake-Science Gap is the difference. Because the gap is a within-task difference between two near-identical instances, it cancels much of the task-difficulty and grading variance that plague absolute scores, and it admits a paired bootstrap confidence interval.

Three design choices make SciHarnessBench scalable, durable, and trustworthy. (i) It is **self-generating and self-grading**: tasks are procedurally instantiated from human-authored templates and graded by deterministic code, with ground truth computed by real scientific libraries (RDKit, SciPy, scikit-learn, Biopython, Astropy). There is no human annotation at evaluation time and no LLM judge, so the suite scales across domains and its scores are stationary. (ii) It is **contamination-resistant**: instances regenerate under a private salt, so a memorized public set confers no advantage and a release can be refreshed rather than rebuilt. (iii) It is **gaming-resistant**: the agent sees an opaque task identifier

(never the variant or seed), detection requires a structured flaw report whose evidence is verified against ground truth, and a flaw claim or abstention on a clean task is a false alarm that scores zero—so “always flag” and “always abstain” strategies destroy competence rather than inflate robustness.

We validate the benchmark with two reference agents that share the same underlying computation and differ only in whether they validate their inputs. The careful agent attains 100.0 % competence and 100.0 % robustness (gap 0.0 pts) without ever false-alarming on a clean task; the naive agent attains 100.0 % competence but 0.0 % robustness (gap 100.0 pts), confirming that the tasks are solvable, the traps are not detectable by the trusting strategy, and the metric separates the two with a wide margin. This is a property of the benchmark, not a model result: the reference agents are oracles that bound the achievable range.

Our contributions are: (1) *fake science* as a measurable construct and the `clean/trapped` paired design with the Fake-Science Gap metric; (2) a self-generating, self-grading, contamination- and gaming-resistant harness realizing it across 8 scientific domains and 12 failure modes; (3) a taxonomy of machine-detectable scientific failure modes grounded in the meta-science literature; and (4) reference-agent and adversarial validation establishing the benchmark’s discriminative range and its resistance to gaming. We release everything as open source.

2 Related Work

Knowledge and predictive benchmarks. Scientific knowledge benchmarks (Rein et al. 2024; Mirza et al. 2025) and broad evaluation suites (Liang et al. 2023) probe what a model knows. Predictive benchmarks fix a task and score accuracy: molecular properties (Wu et al. 2018), PDE dynamics (Takamoto et al. 2022), weather (Rasp et al. 2024), protein fitness (Notin et al. 2023), and materials stability (Riebesell et al. 2025). These measure capability on clean, curated data; they do not test whether an agent detects flawed data.

Agentic science benchmarks. A growing line evaluates tool-using agents on end-to-end tasks: ScienceAgent-Bench (Chen et al. 2025), MLE-bench (Chan et al. 2025), BixBench (Mitchener et al. 2025), LAB-Bench (Laurent et al. 2024), DiscoveryWorld (Jansen et al. 2024), and the AstaBench scientific-agent suite (Bragg et al. 2026), alongside software-engineering agent benchmarks such as SWE-bench (Jimenez et al. 2024). These reward *completing* a workflow and getting a defensible answer. SciHarnessBench is complementary and orthogonal: it holds the workflow fixed and asks whether the agent’s conclusion survives a planted data flaw, scoring the *gap* rather than absolute task success. To our knowledge no prior benchmark uses counterfactual clean/trapped pairs to isolate fake-science failures, nor grades scientific-agent behavior with structured, evidence-verified flaw reports.

Failure modes and meta-science. Our taxonomy operationalizes documented failure modes from the meta-science

literature: the broad unreliability of published findings (Ioannidis 2005), data leakage as a reproducibility threat in ML-based science (Kapoor and Narayanan 2023), false-discovery control under multiple testing (Benjamini and Hochberg 1995), *p*-hacking and researcher degrees of freedom (Simmons, Nelson, and Simonsohn 2011), underpowered studies (Button et al. 2013), circular analysis / double dipping (Kriegeskorte et al. 2009), confounding and Simpson’s paradox (Blyth 1972), batch effects in high-throughput data (Leek et al. 2010), numerical stability conditions (Courant, Friedrichs, and Lewy 1928), and model applicability domains (Tropsha 2010). SciHarnessBench turns each into a machine-checkable trap. Contemporary agents typically reason with chain-of-thought (Wei et al. 2022) and tool use (Yao et al. 2023); whether that reasoning includes the relevant skepticism is precisely what we measure (Zhang et al. 2024).

3 The Fake-Science Problem

We formalize fake science as a property of an agent’s behavior on a pair of instances. A *task family* $\mathcal{F}$ is a procedure that, given a seed s and a variant $v \in \{\text{clean}, \text{trapped}\}$, deterministically produces a task instance. The clean instance x_s^c and the trapped instance x_s^t share an identical base—same underlying system, data, and question—and differ only by a single injected flaw δ in the trapped twin, drawn from the taxonomy of Section 5. We call such a family *paired*; families whose variants necessarily differ in more than δ (e.g. a change in the underlying signal or sample) are marked *non-paired* and reported as separate robustness scenarios, never folded into the gap.

An agent maps a task to an answer, an optional set of flaw reports, and a confidence. For an instance, let a be the indicator that the agent’s answer is the scientifically correct value *for that instance* (the clean answer for a clean task; the flaw-corrected answer for a trapped task), and let d be the indicator that the agent reported the planted flaw with verifiable evidence (Section 6). A trapped instance is handled correctly if the agent either corrects the answer or detects the flaw ($a \vee d$); the dangerous outcome—*fake science*—is a wrong answer asserted confidently and without detection. For a family with N seeds,

$$\text{Competence } C = \frac{1}{N} \sum_s \mathbb{I}[a_s^c], \quad \text{Robustness } R = \frac{1}{N} \sum_s \mathbb{I}[a_s^t \vee d_s^t],$$

and the **Fake-Science Gap** is $\text{FSG} = C - R$. An agent doing real science has $\text{FSG} \approx 0$ (it is as reliable on flawed data as on clean data); an agent doing fake science has high C , low R , and a large positive gap. Reporting C alongside R is essential: a low gap is only meaningful at high competence, and as we show in Section 8 this pairing is what makes the metric immune to trivial gaming.

4 Benchmark Design

The task contract. Each family exposes a generator, a grader, and two reference solvers, all behind a fixed contract. The harness generates an instance, writes its assets into an *opaque* working directory, and hands the agent an

AgentView containing only the public fields: an opaque task identifier, the domain and family, the prompt, the asset files, the requested answer fields, and the controlled vocabulary of flaw kinds it may report. The variant, seed, ground truth, and grading payload are never present in the view. The opaque identifier is a salted hash, so it reveals neither clean/trapped nor the seed, and twins hash to unrelated values.

Structured, evidence-bearing detection. An agent returns answers, a list of structured *issues*—each a (kind, evidence) pair with the kind drawn from the published vocabulary—a boolean abstention, and a confidence in $[0, 1]$. Detection of a trap is credited only if the submission contains an issue of the family’s correct `flaw_kind` whose evidence the grader verifies against ground truth: the exact offending record ids, the detected unit, the non-converged residual, the leaking feature name. Free-text is not searched for keywords. This closes the keyword-stuffing channel: naming the right flaw without the right evidence does not pass.

Reference agents. Two reference solvers per family operate only on the public view. The *naive* solver performs the intended computation but trusts every input and answers with full confidence; the *careful* solver runs the validation a competent scientist would (parse and sanity-check inputs, check units, confirm convergence, recompute from primary data, test for confounds) and either corrects the answer or emits a structured, evidence-bearing issue. These are not systems under test; they are oracles that bound the achievable range and document, per family, what fake-science versus real-science behavior looks like. A model under test is supplied through an adapter that receives only the same public view.

Self-generation and grading. Ground truth is computed at generation time by the relevant scientific library and stored hidden on the instance; the grader compares the submission to it and verifies any flaw report. No human annotates instances and no model judges them, so generation and grading are cheap, deterministic, and reproducible. The cost is human *template* authoring—one generator, grader, and pair of reference solvers per family—which amortizes over unlimited instances.

Contamination resistance and isolation. A private salt mixes into every generator and into the opaque identifiers, so the public, salt-free instances form a transparent development split while official scoring uses a held-out salt and seed range; a release is refreshed by rotating the salt rather than rebuilding the suite. Official evaluation additionally runs the agent out-of-process with only serialized public assets, so the benchmark’s internals (generators, graders, reference solvers) are unavailable to the agent. We detail the threat model and protocol in the supplementary material.

5 Trap Taxonomy

SciHarnessBench plants flaws from 12 failure modes, each chosen to be (a) a documented scientific error and (b)

machine-detectable—a grader can decide, without a human, whether the agent fell for it. They span data trust (decoy values, corrupt or invalid inputs), quantitative reasoning (unit mismatches, out-of-domain extrapolation), numerical practice (non-convergence reported as a result), and statistical methodology (wrong control, confounding/Simpson’s paradox, train/test or target leakage, uncorrected multiple comparisons, circular analysis, underpowered conclusions). Each maps to a single `flaw_kind` that a correct detection must name. A clean task contains no flaw; its trapped twin injects exactly one. Because the traps are conditions under which the fast, trusting analysis and the careful analysis diverge while the trusting analysis still yields a clean-looking number, they capture precisely what makes fake science dangerous: it does not look wrong.

6 Metrics

We report, over paired families, **competence** C and **robustness** R as defined in Section 3; the **Fake-Science Gap** $C - R$ with a **paired bootstrap 95% CI** obtained by resampling (clean, trapped) twins by (family, seed); the **confident-wrong rate** (trapped tasks answered wrong, undetected, with confidence ≥ 0.5 —the directly observed fake-science event); the **false-alarm rate** (clean tasks on which the agent raised an issue or abstained); and the **trap-detection rate** (trapped tasks flagged with a correct, evidence-verified issue). Every metric is reported both micro (pooled) and macro (mean over families), because the suite is trap-imbalanced. Non-paired families are reported as separate robustness scenarios.

The grading policy makes the metric hard to game. On a clean task, the only way to pass is to deliver the correct answer with no flaw claim and no abstention; a false alarm scores zero. On a trapped task, passing requires the corrected answer or a verified structured detection; a bare abstention or a wrong-kind/unverifiable flag does not count. A crash on malformed input is recorded as an unsafe failure (score zero) but, unlike a confident wrong answer, is *not* counted as fake science—these are distinct outcomes a benchmark should not conflate. Because false alarms fail clean tasks, an agent cannot inflate robustness by flagging or abstaining everywhere without destroying its competence; the only route to a low gap is to be genuinely competent and robust.

7 Benchmark Composition

Table 1 summarizes v1: 26 families across 8 domains, exercising all 12 trap types, with 23 paired families and 3 non-paired robustness scenarios. Every family is grounded in a real library, so its ground truth reflects actual scientific computation rather than a hand-coded answer key. Each instance also carries a neutral *cued* prompt and an *uncued* variant with no trap vocabulary, enabling separate reporting of cued and uncued performance.

8 Reference-Agent Validation

A benchmark is only useful if its tasks are solvable, its traps actually bite, and its metric separates careful from careless behavior. We establish all three with the reference agents,

Domain	Families	Trap types exercised
astronomy	3	corrupt input, non-convergence, unit mismatch
biology	3	confounding, corrupt input, wrong control
chemistry	4	corrupt input, decoy data, non-convergence, unit mismatch
climate	3	decoy data, future leakage, unit mismatch
materials	3	corrupt input, extrapolation, unit mismatch
neuroscience	3	circular analysis, multiple comparisons, underpowered
physics	3	extrapolation, non-convergence, unit mismatch
statistics	4	confounding, data leakage, multiple comparisons,
Total	26	12 distinct trap types

Table 1: Composition of SciHarnessBench v1: domains, family counts, and the trap types each domain exercises.

Reference agent	Comp.	Robust.	Gap	Conf.-wrong	False-alarm	Trap det.
naive (trusts inputs)	100.0	0.0	100.0	92.4	0.0	0.0
careful (validates)	100.0	100.0	0.0	0.0	0.0	100.0

Table 2: Reference-agent bounds (%). The two oracles share the same computation and differ only in input validation, bracketing the achievable range. “Gap” is the Fake-Science Gap $C - R$.

run over 12 seeds per family (624 graded tasks each). The results (Table 2, Figure 1) show the intended structure: both agents are fully competent (100.0 % / 100.0 %), so the science is doable; the careful agent is fully robust (100.0 %) with zero false alarms and 100.0 % trap detection; and the naive agent collapses to 0.0 % robustness, with 92.4 % of its trapped tasks lost to confident-wrong answers and the remainder to crashes on corrupt input. The Fake-Science Gap separates them by 100.0 points (naive) versus 0.0 (careful), with a tight paired bootstrap CI of [100.0, 100.0] for the naive agent. The separation is uniform across domains (Figure 2).

Resistance to gaming. We verify the metric cannot be cheated by the strategies that defeat naive benchmark designs. An agent that stuffs every flaw kind onto every task, and an agent that abstains on every task, both collapse to near-zero competence, because their flags and abstentions are false alarms on the clean twins. An agent that attempts to read the variant off the task identifier finds only an opaque hash. An agent that names the correct flaw kind but supplies fabricated evidence is not credited with detection. These are encoded as adversarial regression tests that must continue to fail, alongside per-family discrimination tests and a property test that no clean/trapped label or seed appears in any view or file path.

9 Limitations

SciHarnessBench is procedurally generated and grounded in real computation rather than lifted verbatim from specific papers; the methods and failure modes come from the literature, but the instances are synthetic so they can be regenerated and graded without a human. The reference agents are oracles, not contestants—they bound the discriminative

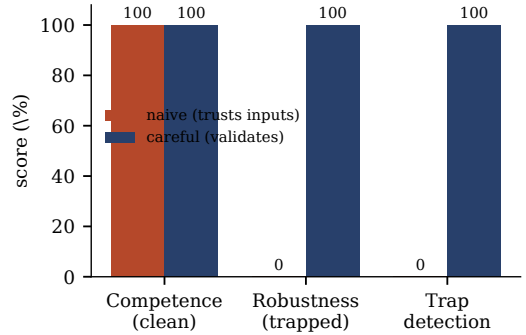


Figure 1: The benchmark’s dynamic range. Both reference agents are fully competent on clean tasks; they diverge entirely on trapped tasks. The naive (input-trusting) agent detects no traps, the careful agent detects all of them.

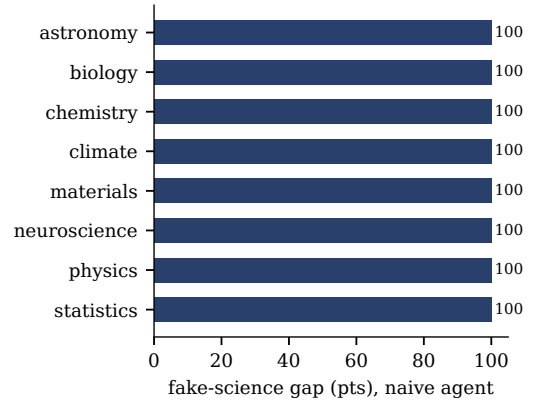


Figure 2: Fake-Science Gap of the naive reference agent by domain. The trap effect is large and uniform across all 8 domains; the careful agent’s gap is zero everywhere (omitted).

range and document correct behavior, and the headline numbers are properties of the benchmark, not of any deployed system. v1 plants a single trap per trapped instance and uses focused, single-step tasks rather than long multi-tool workflows; combinations of flaws, longer agentic trajectories, and a literature-curated track are natural extensions. Finally, this release validates the construct with reference oracles; large-scale evaluation of frontier agents is the subject of ongoing work (Section 10).

10 Model-Evaluation Protocol

We deliberately separate *building and validating* the benchmark from *evaluating* systems on it. Frontier model and agent results are not reported here; they will populate a public leaderboard under the protocol fixed in this paper. Each system is run out-of-process on the private salted split with only serialized public assets, under per-task time and memory limits, through the `(view) → (answers, issues, confidence)` adapter. We will report, for each system, competence, robustness, the Fake-Science Gap with its CI,

the confident-wrong rate, and the false-alarm rate, broken down by domain and trap type, in both cued and uncued tracks, alongside human-analyst baselines. Pre-registering the protocol before evaluation guards against benchmark-specific tuning.

11 Conclusion

Trustworthy scientific automation requires agents that are skeptical of their data, not merely fluent at computing on it. SciHarnessBench makes that skepticism measurable: by pairing each task with a counterfactual twin that hides a single documented flaw, it isolates fake science—confident, well-formatted, wrong results—and quantifies it with the Fake-Science Gap. The benchmark is self-generating, self-grading, contamination-resistant, and gaming-resistant by construction, and its reference agents establish a wide, well-separated dynamic range across 8 domains and 12 failure modes. We hope it provides a durable target for measuring—and improving—the reliability of AI scientists.

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